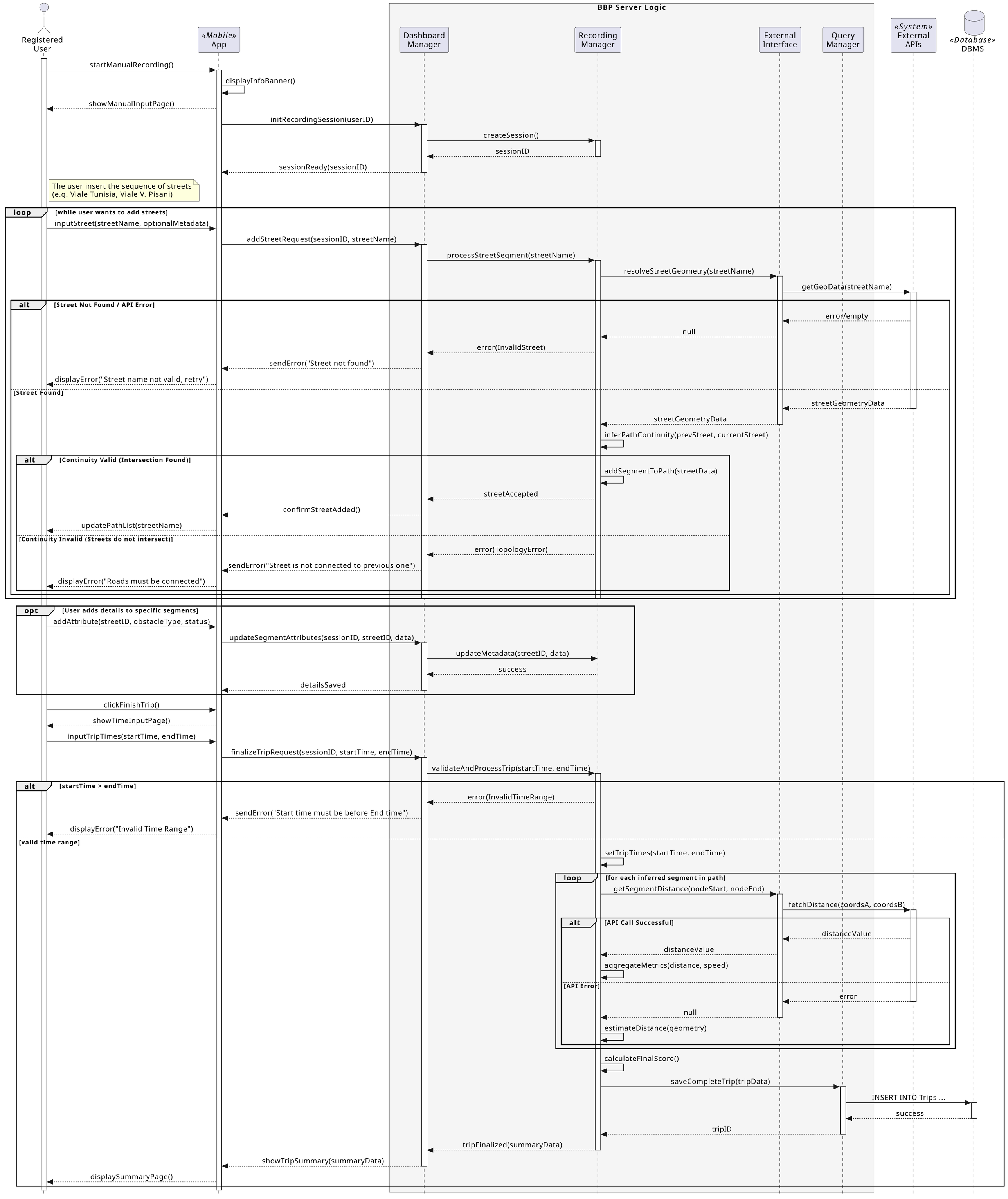
BBP Server Logic
Registered User
«Mobile» App
Dashboard Manager
Recording Manager
External Interface
Query Manager
«System» External APIs
«Database» DBMS
startManualRecording()
displayInfoBanner()
showManualInputPage()
initRecordingSession(userID)
createSession()
sessionID
sessionReady(sessionID)
The user insert the sequence of streets (e.g. Viale Tunisia, Viale V. Pisani)
loop
[while user wants to add streets]
inputStreet(streetName, optionalMetadata)
addStreetRequest(sessionID, streetName)
processStreetSegment(streetName)
resolveStreetGeometry(streetName)
getGeoData(streetName)
alt
[Street Not Found / API Error]
error/empty
null
error(InvalidStreet)
sendError("Street not found")
displayError("Street name not valid, retry")
[Street Found]
streetGeometryData
streetGeometryData
inferPathContinuity(prevStreet, currentStreet)
alt
[Continuity Valid (Intersection Found)]
addSegmentToPath(streetData)
streetAccepted
confirmStreetAdded()
updatePathList(streetName)
[Continuity Invalid (Streets do not intersect)]
error(TopologyError)
sendError("Street is not connected to previous one")
displayError("Roads must be connected")
opt
[User adds details to specific segments]
addAttribute(streetID, obstacleType, status)
updateSegmentAttributes(sessionID, streetID, data)
updateMetadata(streetID, data)
success
detailsSaved
clickFinishTrip()
showTimeInputPage()
inputTripTimes(startTime, endTime)
finalizeTripRequest(sessionID, startTime, endTime)
validateAndProcessTrip(startTime, endTime)
alt
[startTime > endTime]
error(InvalidTimeRange)
sendError("Start time must be before End time")
displayError("Invalid Time Range")
[valid time range]
setTripTimes(startTime, endTime)
loop
[for each inferred segment in path]
getSegmentDistance(nodeStart, nodeEnd)
fetchDistance(coordsA, coordsB)
alt
[API Call Successful]
distanceValue
distanceValue
aggregateMetrics(distance, speed)
[API Error]
error
null
estimateDistance(geometry)
calculateFinalScore()
saveCompleteTrip(tripData)
INSERT INTO Trips ...
success
tripID
tripFinalized(summaryData)
showTripSummary(summaryData)
displaySummaryPage()